

Optimized Attention-Enhanced Physics-Guided Neural Network for Satellite-Based Ocean Subsurface Temperature Predicting

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Abstract—Understanding subsurface ocean properties is fundamental to advancing our knowledge of ocean dynamics. However, the scarcity of in situ observations imposes significant limitations on comprehending underwater structures. Satellite data, with its broad spatial coverage and temporal frequency, offer valuable insight to estimate subsurface conditions from surface observations. This study introduces a novel deep learning method aimed at improving the accuracy and physical plausibility of ocean subsurface temperature (ST) predictions. By integrating knowledge of ocean stratification, attention mechanisms, and a multiloss function optimization approach, the model facilitates a robust integration of data-driven methods with theoretical principles. Experimental results demonstrate a performance enhancement of 20% compared to a convolutional neural network (CNN) and 4% compared to UNet, alongside significant improvements in physical plausibility. This work not only provides a reliable approach for ST prediction but also highlights the potential of integrating physical knowledge into deep learning frameworks, paving the way for more accurate and physically guided ocean modeling.

Index Terms—Multiloss optimization, physical plausibility, physics-guided deep learning, satellite-based prediction, subsurface temperature (ST).

I. INTRODUCTION

IN THE era of climate change and ocean warming, the investigation of the ocean's subsurface temperature (ST) has become increasingly critical. ST is not only a key variable in marine ecosystems and global climate systems, but also has important practical significance for fisheries, navigation, and

marine resource development. A deeper exploration into the ocean's subsurface physical properties is vital for elucidating its internal dynamics [1]. Nevertheless, due to the sparsity and incompleteness of underwater observation, accurately predicting ST remains a huge challenge.

Advancements in remote sensing technology have revolutionized oceanic observation by providing an increasing amount of high-resolution and long-term data. However, they are limited to the surface layer [2]. Progress in deep learning has enhanced the captioning of underwater image features. Novel approaches, such as attention mechanism-based fusion and AquaSketch-Enhanced Cross-Scale Information Fusion, play a crucial role in applications, such as ocean geoscience and underwater remote sensing [3], [4]. Satellite observations do not directly represent underground information, but certain aspects of underground structures can be inferred from surface data [5], [6]. Previous studies have explored the feasibility of constructing subsurface structures from surface information [7], [8].

Satellite-based reconstruction of ocean surface parameters has demonstrated robust performance through the use of traditional machine learning algorithms, including support vector machines (SVM) [9], Random Forests (RF) [10], Extreme Gradient Boosting (XGBoost) [11], and Extremely Randomized Trees (Extra-Trees) [12]. Ensemble methods also extend to predict ST [1]. However, ensemble models grapple with spatial-temporal complexity and high-dimensional information, often requiring extensive training datasets and supplementary algorithmic support [13], [14]. The main reason is that although ensemble algorithms enhance generalization ability by combining the results of multiple weak learners, they still fail to fully explore the semantic features of 3-D data [6]. Compared to traditional machine learning algorithms, deep learning has excelled in capturing higher level spatiotemporal features in earth science [15], enhancing the accuracy of ST prediction, studies including artificial neural networks (ANN) [16], convolutional neural networks (CNN) [6], [17], UNet [18], and ConvLSTM [19]. Notable advances, such as knowledge-informed Clustering-Guided and Knowledge-Distillation Network (CGKDN), have enhanced the interpretability and accuracy of subsurface estimation, with clustering and knowledge distillation from global to local patterns fusing [20].

Despite the progress made by traditional machine learning and deep learning methods in predicting subsurface ocean temperature, several critical limitations remain. First, due to the

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high cost and sparse distribution of in situ ocean observations, particularly in deep-sea and remote areas, the number of available samples is limited. This data sparsity makes it difficult for models to fully learn the spatial variability of subsurface structures, thereby reducing the comprehensiveness and generalizability of predictions. Second, traditional machine learning and deep learning methods exhibit limited capability in modeling the spatiotemporal relationships among multisource remote sensing variables, such as sea surface temperature (SST), sea level anomaly (SLA), wind, and salinity. This often leads to suboptimal fusion performance and insufficient accuracy in reconstructing 3-D temperature structures. Furthermore, reliance solely on data-driven approaches may conflict with established physical oceanography principles, and research on physics-guided deep learning has emerged in recent years [21]. Incorporating physical constraints and empirical knowledge within these models is pivotal for minimizing physical inconsistency and enhancing prediction reliability [22], [23], [24]. This integration significantly improves the fidelity and applicability of predictive models in oceanographic research.

These challenges have not yet been effectively addressed in existing works, making it imperative to develop models that can incorporate physical consistency, adapt to sparse observational data, and effectively fuse multisource information. Building on advances in predicting ST, this study proposes a novel 3-D Multiple-objective Physics-Guided Convolutional Block Attention Neural Network (3D-MOPGCBANN) to refine the accuracy of ST predictions. First, a physics-guided module is developed to integrate physical models with observational data, enhancing the understanding of marine physics and improving predictive accuracy. Second, the model incorporates spatial and channel attention mechanisms within its convolutional block, allowing for the prioritization of critical features and more precise tracking of ST distributions in complex marine settings. Then a multiobjective optimization (MO) module is designed due to physics-guided loss adjusting, refining model's performance. The main contributions of this article are as follows.

- 1) A neural network with physical knowledge constraint modules is designed to predict ST. By embedding the physical laws of ocean stratification stability for all depth levels into the loss function, the PG module can identify outputs that do not conform to this law and update the model parameters by backpropagation of the added loss function. This improves the model's physical rationality and enhances its generalization ability.
- 2) To further enhance the information extraction ability of the model, the construction of spatial and channel attention allows the model to selectively focus on salient satellite surface observations that influence ST at the target depth, thereby capturing complex temperature distribution more effectively and improving the accuracy of the model.
- 3) Due to the fact that physical constraints are typically embedded into loss functions, model training requires balancing between physics loss and regression loss. The traditional approach is to perform a weight-step search across multiple losses. This study designed an MO module, guided by the Karush–Kuhn–Tucker (KKT)

conditions, to optimize multiple loss functions in physical neural networks. This module efficiently manages trade-offs among different loss functions, enhancing optimization speed and effectiveness. The remainder of this article is structured as follows. Section II reviews traditional machine learning and deep learning methods of temperature reconstruction, physical knowledge incorporation, and MO research. Section III elaborates on the 3D-MOPGCBANN model. Section IV describes the experimental design of datasets and preprocessing techniques. Section V presents experimental results, comparisons, and ablation studies. Section VI concludes with a summary of the findings and the study's broader implications.

II. RELATED WORKS

A. Traditional and Deep Learning of ST Prediction

Predicting ST is a critical yet challenging task in marine sciences, exacerbated by sparse direct oceanic measurements [1]. Historically, physical and empirical statistical models have been staples in this domain, employing numerical modeling to capture oceanic temperature and salinity variations [25]. The integration of enhanced observational inputs, such as satellite data, Argo floats, and reanalysis assimilation products with advanced artificial intelligence, has revolutionized the reconstruction and prediction of the ocean's 3-D thermal structure. This synergy provides vital data for studying oceanic processes and climate change impacts.

Among machine learning techniques, ensemble methods like RF, LightGBM, and XGBoost have shown promise in estimating ST by improving model generalization through the combination of multiple weak learners [1], [13], [14]. Despite their effectiveness, these models cannot generally autonomously extract deep spatial-temporal features, as they depend on predefined feature representations. Conversely, deep learning methods such as CNN and long short-term memory networks (LSTM), along with advanced architectures like UNet and ConvLSTM, have excelled by independently extracting intricate spatial-temporal features, significantly enhancing prediction accuracy [6], [16], [17], [18], [19]. These models adeptly learn from extensive datasets, enabling them to predict ST distributions with high precision, thus providing a more dynamic and detailed understanding of the ocean's internal processes.

B. Physics Knowledge Guided Model

Recent strides in machine learning have prominently featured Physics-Informed Neural Networks (PINNs), which incorporate physical laws as foundational knowledge for solving both forward and inverse problems involving partial differential equations (PDEs). This approach allows the embedding of physical dynamics directly into neural networks, as demonstrated by Raissi et al. [26], who utilized this method to simulate fluid dynamics parameters such as velocity and pressure. De Bézenac et al. [27] illustrated the enhancement of model performance in predicting complex natural phenomena like SST through the integration of physical knowledge into deep learning architectures. Furthermore, Seo et al. [28]

extended the application of physical laws into graph neural networks to predict temperatures.

In the realm of ocean temperature forecasting, an intricate understanding of physical mechanisms and empirical models proves crucial for reducing prediction errors that do not conform to known physical laws, thereby improving the accuracy of models. Wu et al. [24] emphasized the significance of incorporating physical information about internal ocean waves into models, markedly enhancing their predictive capability. In a parallel development, Meng et al. [23] showcased how leveraging physical mechanisms through a generative adversarial network (GAN) model, pretraining a discriminator within HYCOM data, can significantly boost model performance. Liu et al. [29] took 1-D physical eddy parameters as prior knowledge-embedded DNN branch, to explore thermohaline structures of mesoscale eddies. These examples underline the growing importance and effectiveness of incorporating scientific knowledge into computational frameworks to enhance the fidelity and robustness of predictive models in marine science.

C. MO in Neural Network

MO theory is instrumental in handling complex systems characterized by competing objectives, providing a sophisticated framework to discover optimal solutions [30], [31]. In the realm of PINNs, the integration of physical knowledge necessitates a delicate balance among multiple loss functions to ensure both high data fidelity and strict adherence to physical laws. Real-world data, whether observational or assimilated, typically introduces additional complexity due to its noisy nature. This complexity often creates inherent conflicts between data-driven and physics-driven losses, requiring a refined strategy to achieve an optimal equilibrium [32]. Lin et al. [33] addressed this challenge by leveraging the KKT conditions, developing a two-step Pareto efficient optimization algorithm specifically tailored for MO. Originally applied within e-commerce recommendation systems, this approach has proven beneficial in optimizing multiple objectives within physics-guided neural networks, ensuring physical consistency while effectively balancing multiple competing objectives.

Further exploring the dynamics of MO in PINNs, Rohrhofer et al. [32] investigated the use of diffusion and Navier–Stokes equations to assess how system parameters influence the shape of the Pareto front. Their research also evaluated the effectiveness of adaptive activation functions and adaptive loss weighting techniques, providing deeper insight into optimizing PINNs. This comprehensive approach underscores the crucial role of MO in enhancing the robustness and efficacy of physics-guided models, particularly in complex and noisy environments.

III. METHOD

A. Overview of 3D-MOPGCBANN Model

The 3D-MOPGCBANN model utilizes various satellite variables to predict ST. These input variables include SST, sea surface salinity (SSS), SLA, and zonal and meridional wind speeds (SSW). Specifically, the model is comprised of three modules. First, to enhance the extraction of spatial features from sea surface data, spatial attention mechanisms and channel attention modules (CBAM) are integrated into the CNN

architecture. Second, a physical knowledge guided (PG) module is proposed, as shown in Fig. 1. This module incorporates a physical loss function along with data loss, which guides the training of the model's parameters to minimize physical error. This module aims to mitigate predictions inconsistent with physical equations, thereby enhancing model accuracy. Third, to address the optimization challenge posed by multiple loss functions following the inclusion of a physically guided loss function, an improved MO algorithm for embedding loss functions is employed. This algorithm optimizes multiple loss functions, achieving a Pareto frontier that enhances the weight balance and training speed of the physics neural network. All these three components construct the proposed 3D-MOPGCBANN.

B. Attention-Enhanced CNN Module

Convolutional block attention module (CBAM) is embedded into the model to enhance its capability in extracting spatial regions and features relevant to subsurface temperature [34]. The attention mechanism, proposed by Vaswani et al. [35], mimics human attention mechanisms and has yielded significant advancements across various domains in deep learning. Spatial attention mechanisms and channel attention mechanisms represent extensions of this mechanism, aiming to augment the model's sensitivity to specific features in input data [36], [37], as shown in Fig. 2.

Spatial attention mechanism aims to enable models to focus on the most crucial spatial regions within the input data, and channel attention mechanism focuses on multiple channels of input data, applying channel weights to highlight important information. The integration of spatial and channel attention mechanisms significantly enhances the deep learning model's capabilities, leading to marked performance improvements in image recognition, classification, and other visual tasks. The synergistic combination of these mechanisms allows models to process and parse input data with greater precision, offering robust support for the understanding and prediction of subsurface complex scenes. The equations of CBAM are as follows:

$$X_{\text{out}}^C = \sigma(X_{\text{avg}}^C + X_{\text{max}}^C) \cdot X_{\text{in}} \quad (1)$$

$$X_{\text{out}}^{\text{CS}} = \sigma(X_{\text{avg}}^{\text{CS}} + X_{\text{max}}^{\text{CS}}) \cdot X_{\text{in}}^C \quad (2)$$

where X_{out}^C denotes the output of channel attention module, and X_{avg}^C and X_{max}^C represent the average and maximum of X in channel dimension, respectively. $X_{\text{out}}^{\text{CS}}$ denotes the output of spatial attention module based on channel module, and $X_{\text{avg}}^{\text{CS}}$ and $X_{\text{max}}^{\text{CS}}$ represent the average and maximum of X^C in spatial dimension.

C. Physics-Guided Module

Considering that the vertical distribution of seawater temperature is influenced by ocean stratification, this study integrates the ocean stratification stability model as prior physical knowledge into neural networks. The upper ocean exhibits lower average density, while the deeper layers have higher average density, resulting in a relatively stable vertical thermohaline structure [38], [39]. This structure manifests as a hierarchical distribution of thermodynamic state parameters such

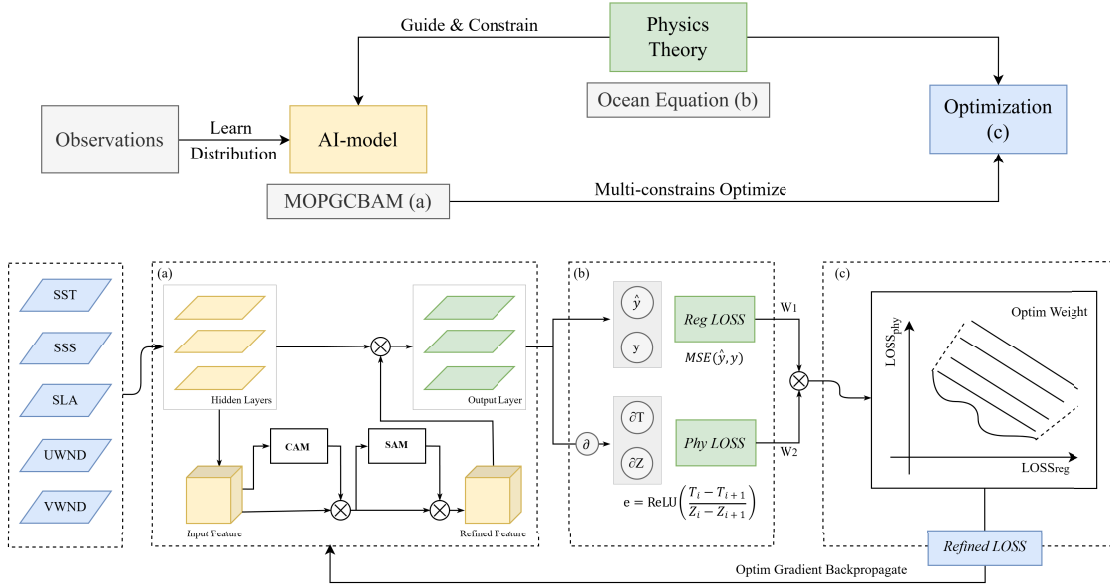


Fig. 1. Overview of the proposed 3D-MOPGCBANN model. (a) Attention-enhanced CNN module extracts multi-source sea-surface signals (SST, SSS, SLA, U/V-wind) with a CNN enhanced by channel and spatial attention. It learns representative features and produces the initial subsurface temperature estimate used by later modules. (b) Physics-guided module injects prior knowledge of vertical temperature gradients to regularize the network. It computes a physics-consistency penalty and combines it with the data regression loss to guide training, suppressing physically inconsistent predictions. (c) Multiloss optimization Module balances the two objectives (data fit and physics consistency) by adaptively adjusting their weights based on KKT/Pareto principles, yielding a refined overall loss and faster, more stable convergence.

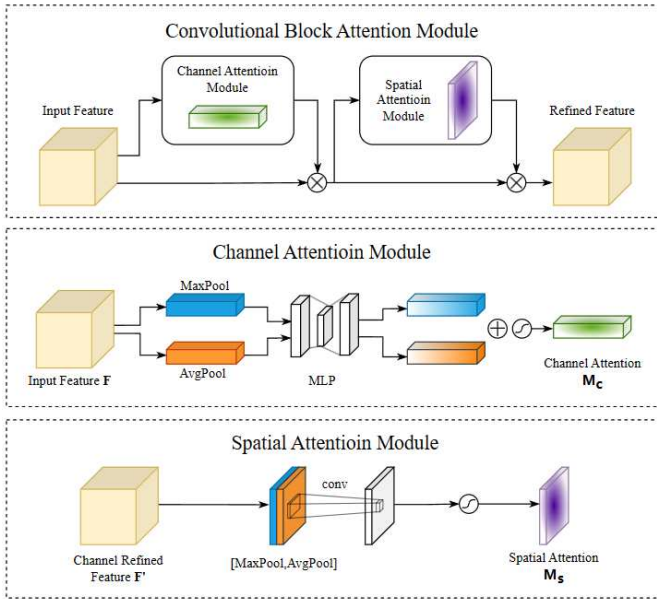


Fig. 2. Details of CBAM block.

as density, temperature, and salinity with depth, which has become increasingly pronounced over the past half-century [40]. In oceanography, the stability condition formula for ocean stratification is used to approximate the characteristics of stable stratification [41]

$$N^2 = g \left(-\frac{1}{\rho} \right) \frac{d\rho}{dz} \quad (3)$$

where the parameters $d\rho$, dz , and g denote the seawater density gradient, depth gradient, and gravitational acceleration, respectively. Buoyancy frequency serves as a measure of the fluid's

stability against vertical displacements. Greater stratification in the water column corresponds to increased static stability and a higher buoyancy frequency. The squared buoyancy frequency, N^2 , is calculated based on the Thermodynamic Equation of Seawater 2010 (TEOS-10) [42]

$$N^2(\Theta, S_A) = g(\alpha^\Theta \Theta_z - \beta^\Theta S_{Az}) \quad (4)$$

where α^Θ and β^Θ represent the thermal expansion and saline contraction coefficients, respectively. Subscript (z) indicates the vertical gradients. Assuming that the salinity gradient within the study area is not substantial, the stability of stratification can be characterized by the temperature profile. The stability condition for stratification effectively reduces to the following expression:

$$\frac{dT}{dz} > 0 \quad (5)$$

where T denotes the temperature, and z represents the depth. Consequently, the rate of temperature change between discrete depths, or the temperature gradient, can be calculated. This term is incorporated as a loss function into the propagation of model parameters, thereby employing prior knowledge to guide the training of the model on stratification stability. The loss function is expressed as follows:

$$\text{LOSS}_{\text{phy}} = \text{ReLU} \left(\frac{T_i - T_{i+1}}{Z_i - Z_{i+1}} \right). \quad (6)$$

D. Multiloss Optimization Module

Physical loss of ocean stratification is used as a model constraint, necessitating a balance between data and physical loss functions during model training. Traditional approaches involve weight tuning across multiple losses, while this study designed an MO module. MO module typically deals with

multiple conflicting objective functions, aiming to find a set of solutions that achieve the best trade-offs among different objectives, known as the Pareto front [43]. Any solution on the Pareto front cannot improve in one objective without deteriorating in at least one other objective.

Within the framework of nonlinear programming, the KKT conditions serve as a pivotal method for ascertaining optimal solutions. They constitute a set of necessary conditions (and under certain circumstances, sufficient conditions) for validating whether a local optimum qualifies as a global optimum. Pertinent to MO, the KKT conditions facilitate the identification of solutions that conform to Pareto optimality [33]. In the context of 3D-MOPGCBANN training, the application of the KKT conditions to achieve Pareto Front optimization involves the following steps.

Construction of the objective function: 3D-MOPGCBANN is characterized by two loss functions, designated to reflect data fitting and physical knowledge losses, respectively. The loss function based on physical knowledge has been defined in (6). Losses based on data fitting are defined by the following formula:

$$\text{LOSS}_{\text{reg}} = \text{mse}(\hat{y}, y) \quad (7)$$

where y represents the labels, while $\hat{y}$ denotes the outputs of the model. Two loss functions are aggregated into a composite objective function via a set of continuously iteratively updated weights.

$$\text{LOSS}_{\text{total}} = w_1 * \text{LOSS}_{\text{phy}} + w_2 * \text{LOSS}_{\text{reg}} \quad (8)$$

where w_1 and w_2 corresponds to the current weights assigned to LOSS_{phy} and LOSS_{reg} , respectively. Integrating these loss functions, the model parameters are updated through two-stage algorithms within each training batch.

In the first stage, both loss functions are computed based on the discrepancy between the model outputs and labels. Gradients concerning the model parameters are derived for each loss function, yielding the following gradient expressions:

$$\text{Grad}_i = \frac{\partial L(\theta)}{\partial \theta} = w_1 \cdot \left(\text{ReLU} \left(\frac{\partial \hat{y}}{\partial z} \right) \right) + w_2 \cdot (\hat{y} - y) \frac{\partial \hat{y}}{\partial \theta} \quad (9)$$

where z denotes the value of predict depth, and θ denotes all parameters of the model. Predicated on the KKT conditions, a Pareto solution for the multiobjective scenario can be attained when the model parameters fulfill the following criteria:

$$\begin{aligned} \sum_{i=1}^K \omega_i &= 1, \quad \exists \omega_i \geq c_i, \quad i \in \{1, \dots, K\} \quad \text{and} \\ \sum_{i=1}^K \omega_i \nabla_{\theta} \mathcal{L}_i(\theta) &= 0. \end{aligned} \quad (10)$$

Here our optimization objective transitions to locating a set of w_i that minimizes the following objective function, LOSS_{phy} and LOSS_{reg} , thereby approximating the KKT conditions:

$$\min. \left\| \sum_{i=1}^K w_i \nabla_{\theta} \mathcal{L}_i(\theta) \right\|_2^2 \quad (11)$$

$$\text{s.t.} \quad \sum_{i=1}^K w_i = 1, \quad w_i \geq c_i \quad \forall i \in \{1, \dots, K\} \quad (12)$$

where w_i is the weight of LOSS_i , c_i is the lower limit of w_i , K is the number of w_i , and $\nabla_{\theta} \mathcal{L}_i(\theta)$ denotes the Gradients _{i} . To address this objective, the Lagrangian multiplier is employed to deduce an optimal set of solutions for w_i that satisfy the established conditions, as follows:

$$W = \left((M^T M)^{-1} M \tilde{z} \right) [1:K] + C \quad (13)$$

$$M \text{ is } \begin{bmatrix} GG^T & \mathbf{e} \\ \mathbf{e}^T & 0 \end{bmatrix} \quad (14)$$

where W is the vectorized w_i , C is the vectorized c_i , $\tilde{z}$ is the concatenated vector of $-GG^T C$ and $1 - \sum C$, and G is the gradient matrix of $\nabla_{\theta} \mathcal{L}_i(\theta)$.

In the second stage, leveraging the compute w_i , the $\text{LOSS}_{\text{total}}$ for the current batch can be calculated by (8) and backpropagation is performed on $\text{LOSS}_{\text{total}}$

$$\theta' = \theta - \text{Grad}(\text{LOSS}_{\text{total}}, \theta) * \text{lr} \quad (15)$$

where θ is the origin model parameters, θ' is the updated model parameters, and lr is the current learning rate.

The flowchart of the MO model is shown in Fig. 3, through which the 3D-MOPGCBANN model is capable of effectively balancing between the data fidelity and the adherence to physical knowledge objectives. This approach improves the traditional exhaustive search for weight parameters, not only accelerating the model optimization process but also achieving optimal solutions across multiple performance metrics. Additionally, it provides theoretical underpinnings for minimizing data discrepancy and ensuring physical consistency within the model.

IV. EXPERIMENTS DESIGN

A. Study Area and Data

Tropical Pacific (180–120W, 10–10N, TP) and Northwest Pacific (150–170E, 20–40N, NWP) are selected as our case study locations, as shown in Fig. 4. Those are two typical areas where TP contains a renowned calculation area of Niño 3.4 and NWP is the flow area of Kuroshio Current (also known as the Japan Warm Current), rendering both exemplary areas for research. The data used in this study comprise multisource satellite products and gridded ARGO sea temperature data. The sea temperature data are obtained from the International Pacific Research Center (IPRC), Hawaii, HI, USA, for the gridded Argo data (<http://iprc.soest.hawaii.edu>). This dataset provides a 3-D temperature variable. The spatial resolution of the temperature monthly average product is $1^\circ \times 1^\circ$.

The satellite datasets include SST, SSS, sea surface wind (SSW), and SLA. The SST data are obtained from the NOAA Optimal Interpolated SST (OISST) dataset (<https://www.ncei.noaa.gov/products/optimum-interpolation-sst>), which uses SST products from multisatellite sensors [44], [45], calibrated by observations from different platforms (ships and buoys) [46]. The spatial resolution of SST data is $0.25^\circ \times 0.25^\circ$. SSS data are from the European Space Agency (ESA) Climate Change Initiative (CCI) program (<https://catalogue.ceda.ac.uk/uuid/aea368301bff448bbbea7e90e66819e5>) [47]. The datasets are fused with SMOS and SMAP satellite data and validated against in situ data. In this product, the spatial resolution of the data is 25 km. SSW data is the Cross-Calibrated Multi-Platform (CCMP) L3 V2.0 ocean vector wind

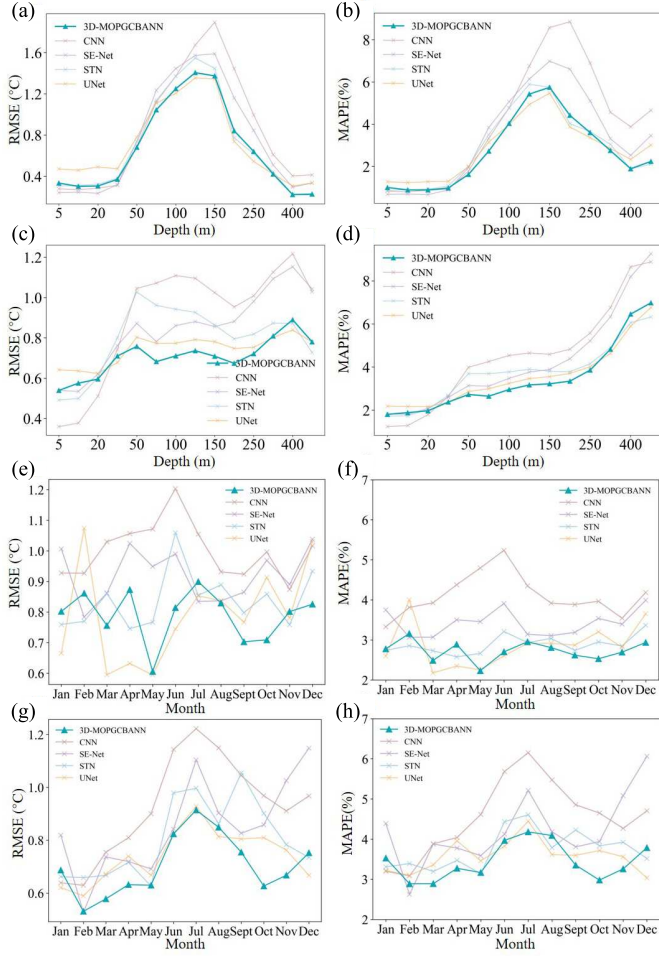


Fig. 3. Flowchart of multi-objective optimization training module. Step I denotes the optimization of weight values. Step II denotes the back-propagation of aggregated optimized loss.

analysis product (<http://www.remss.com/measurements/ccmp/>) [48], using the variational analysis method (VAM) combined with cross-calibrated multisatellite microwave winds and instrumental observations [49]. It has a spatial resolution of $0.25^\circ \times 0.25^\circ$. SLA data is from the Absolute Dynamic Topography products of Archiving, Validation, and Interpretation of Satellite Oceanographic data (AVISO) project (<https://www.aviso.altimetry.fr/en/data/products/sea-surface-height-products/global/>). It has a spatial resolution of $0.25^\circ \times 0.25^\circ$.

The sea temperature data are obtained from the International Pacific Research Center (IPRC), Hawaii, HI, USA, for the gridded Argo data (<http://iprc.soest.hawaii.edu>). This dataset provides a 3-D temperature variable. The spatial resolution of the temperature monthly average product is $1^\circ \times 1^\circ$, and it has 27 vertical layers above 2000 m, including 0 m as the first layer. We chose 26 layers below the sea surface as the label.

B. Data Preprocess

We unify the horizontal grid of these gridded data to $0.5^\circ \times 0.5^\circ$ spatial resolution using the horizontal mapping and interpolation functions provided by the Climate Data Operators computational tool. Within the TP and NWP study regions, the training-set range of ST is from 7.3°C to 30.7°C and from 3.5°C to 30.4°C , respectively. The study period

TABLE I
SETTINGS OF THE EXPERIMENTAL MODELS

Model	Description of Model Setting
RF	N_estimators=200, Max_depth=None
XGBoost	N_estimators=200, Max_depth=None
CNN	8 Blocks (Conv)
SE-Net	8 Blocks (Conv + Channel Attention)
STN	8 Blocks (Conv + Spatial Attention)
UNet	3-level Encoder-Decoder, skip connections, DoubleConv blocks
ConvNeXt	4 Stages, FPN decoder, LayerNorm2d, DropPath=0.1
3D-MOPGCBANN	8 Blocks (Conv + Spatial Attention + Channel Attention)

is from 2010 to 2018, and the temporal scale of this study is set at a monthly interval. Standardization is performed on each variable for neural network training, according to the following equation:

$$x'_i = \frac{x_i - \bar{x}_i}{\sigma_i} \quad (16)$$

where x_i means each variable, $\bar{x}_i$ and σ_i mean the mean value and the standard deviation of each variable, respectively.

C. Model Metrics

The evaluation metrics used in the experiments are RMSE, mean absolute error (MAE), mean absolute percentage error (MAPE), and R -squared (R^2). RMSE indicates the square root of the mean squared difference between the actual and predicted values, indicating the model's goodness of fit. MAE is the mean of the absolute difference between the actual and predicted values, measuring the magnitude of the prediction error. MAPE measures the percentage of the MAE relative to the actual value, independent of the measurement range. R^2 is used to describe how well the data fits the model. The definitions of these metrics are as follows:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (Ym_i - Yp_i)^2} \quad (17)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |Ym_i - Yp_i| \quad (18)$$

$$\text{MAPE} = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{Ym_i - Yp_i}{Ym_i} \right| \quad (19)$$

$$R^2 = \frac{(\sum_{i=1}^n (Ym_i - \bar{Ym}) (Yp_i - \bar{Yp}))^2}{\sum_{i=1}^n (Ym_i - \bar{Ym})^2 \sum_{i=1}^n (Yp_i - \bar{Yp})^2} \quad (20)$$

where n indicates the number of the test set, and Ym and Yp are the measured and predicted ST, respectively.

D. Implementation Details

This section outlines the experimental setup. For a thorough comparative evaluation, we utilize several models, including the conventional CNN, UNet [50], ConvNeXt [51], spatial attention-augmented STN [37], channel attention-enhanced SE-Net [36], and the ensemble model XGBoost [52]. Identical network architectures and hyperparameters are employed across the CNN, SE-Net, STN, and 3D-MOPGCBANN for comparison. XGBoost is trained with default parameters. The UNet model adopts an encoder-decoder structure with skip connections and DoubleConv blocks. ConvNeXt follows a hierarchical design with multiple stages and an FPN decoder. Detailed experimental settings are provided in Table I.

TABLE II
OVERALL MODELS' PERFORMANCE EVALUATION IN PREDICTING ST FROM SATELLITE IMAGES

Type	Model	RMSE/°C	TP area MAE/°C	MAPE/%	RMSE/°C	NWP area MAE/°C	MAPE/%
Reanalysis	SODA	3.891	2.722	14.393	0.859	0.611	3.984
	ORAS	2.342	1.491	8.967	1.785	1.391	9.076
AI Models	ConvNeXt	1.148	0.867	5.298	1.030	0.790	5.025
	RF	1.148	0.637	3.464	1.212	0.844	5.300
	XGBoost	1.065	0.615	3.382	1.142	0.818	5.150
	CNN	1.005	0.679	4.109	0.947	0.711	4.550
	SE-Net	0.924	0.605	3.426	0.867	0.655	4.222
	STN	0.842	0.551	2.888	0.816	0.603	3.736
	UNet	0.805	0.525	2.862	0.748	0.564	3.573
	3D-MOPGCBANN	0.794	0.498	2.733	0.712	0.542	3.447

All models utilize data spanning from 2010 to 2015 as the training set, data from 2016 to 2017 as the validation set, and data from 2018 as the test set. To circumvent overfitting, an early stopping technique is implemented, halting the training process when the model's performance begins to degrade on the validation set. All models are operationalized using PyTorch 1.10 and trained with an RTX 3090 GPU. The training is conducted using Adam optimizer with an initial learning rate set at 10^{-3} . The batch size is configured to 72.

V. EXPERIMENTS RESULT

A. Evaluation on Model Performance

The experimental results offer a comparison of prediction errors between the proposed 3D-MOPGCBANN model and several compared models, demonstrating its effectiveness and applicability in ST forecasting (Table II). Initially, we evaluated the model against advanced global reanalysis datasets, such as Simple Ocean Data Assimilation (SODA) v3.4.2 [53] and Ocean Reanalysis System 5 (ORAS5) [54]. Notably, this model, which relies solely on satellite-derived sea surface data and is trained with ARGO data as reference labels, outperformed reanalysis products that incorporate observational data and oceanic numerical simulations. The model exhibited an average performance improvement of over 45% across two study areas. This enhancement can be attributed to the model's capacity to identify the underlying relationship between subsurface and surface data, leading to superior predictive accuracy. We further compared the proposed model with both traditional ML models and advanced DNNs. The ML models included ensemble methods such as RF and XGBoost, which have previously demonstrated efficacy in predicting ST and density [1], [55]. The DNNs tested included CNN, UNet, ConvNeXt, and attention-enhanced CNN variants, such as STN, SE-Net. The 3D-MOPGCBANN model reduced RMSE, MAE, and MAPE by 32%, 26%, and 26%, respectively, outperforming XGBoost. This enhancement likely arises from the limitations of ensemble ML models, which struggle to capture spatial neighborhood dependencies critical for ST inference [6]. By leveraging convolutional architectures, deep learning effectively integrates spatial correlations, addressing the strong interdependence between surface and adjacent ST. Furthermore, our model demonstrated superior performance compared to the standard CNN model, reducing RMSE, MAE, and MAPE by approximately 23%, 25%, and 29%, respectively. It also reduced the RMSE, MAE, and MAPE errors

compared to the suboptimal model, UNet, by 1.4%, 5.1%, and 4.5% in the TP area, and by 4.8%, 17.1%, and 3.5% in the NWP area. Overall, the proposed 3D-MOPGCBANN model outperforms all benchmarks, recording the lowest errors (RMSE: 0.794 °C in TP, 0.712 °C in NWP), highlighting its enhanced predictive accuracy and robustness in capturing regional 3D temperature dynamics.

B. Optimized Spatial Distribution Across Varying Depths and Months

Section V-A has demonstrated the effectiveness of the 3D-MOPGCBANN model. In this section, we further assessed the model predictive performance across 14 depth layers and 12 months of the 2018 test period (Fig. 5), comparing with the ARGO dataset from the International Pacific Research Center (IPRC). In terms of depth perspective, monthly predictions were averaged to derive annual means for depths of 5, 10, 20, 30, 50, 75, 100, 125, 150, 200, 250, 300, 400, and 500 m. The 3D-MOPGCBANN model outperformed all compared models in over 50% of the depth layers. However, substantial errors were observed in the 50–200 m depth range within the TP area, where rapid thermocline fluctuations hinder model accuracy [Fig. 6(e1)–(j1)]. In contrast, prediction errors in the NWP area increase with latitude and thermocline depth. These errors predominantly grow with depth, as subsurface conditions are inferred solely from surface data. In a temporal perspective, errors in the TP region remained stable across months, with May exhibiting the lowest error peak. In the NWP region, errors exhibited an increasing trend from June to September across all models. Overall, the 3D-MOPGCBANN model demonstrated superior performance across nearly all months. These depth error trends align with the recently proposed CGKDN, where the peak errors of both models appear to be at around 150 m depth level.

The 3D-MOPGCBANN model accurately predicts the spatial distribution of underwater temperature across varying depths using satellite-derived sea surface data. In the TP area, it effectively reproduces the “cold tongue” phenomenon, where cold water flows from east to west at the equator (0° latitude), a feature influenced by ocean currents and climatic conditions. This behavior corresponds to the temperature gradients observed during El Niño-La Niña cycles [56]. In the NWP area, the temperature distribution reflects the influence of the Kuroshio Extension, characterized by higher temperatures to the south and lower temperatures to the north [57]. The

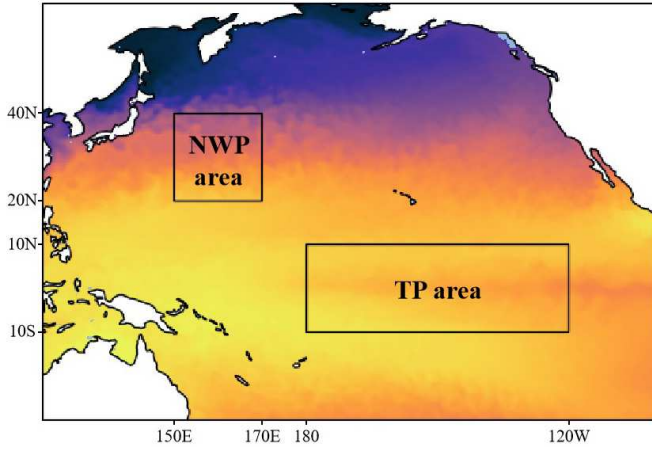


Fig. 4. Study areas of tropical pacific (180W–120W, 10S–10N, TP) and northwest pacific (150E–170E, 20N–40N, NWP).

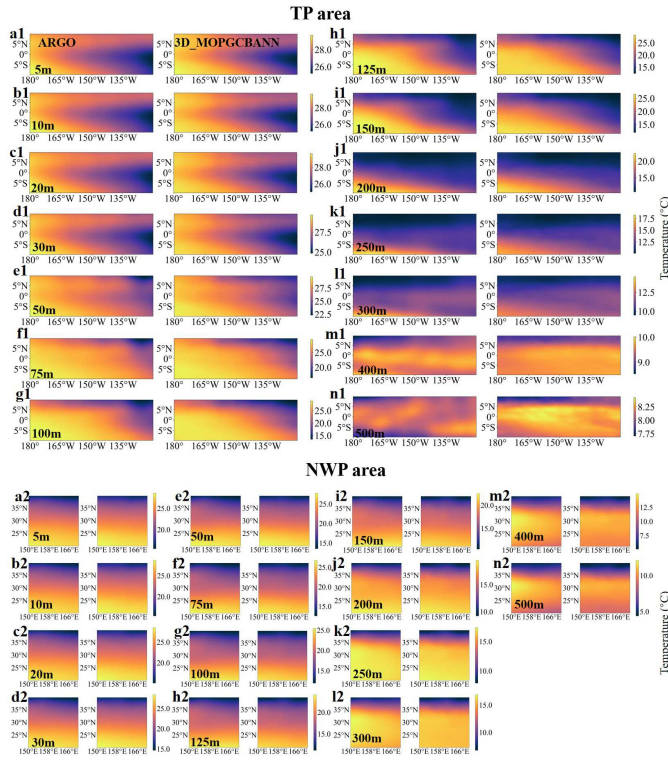


Fig. 5. Evaluation metrics between 3D-MOPGCBANN and comparison models (suboptimal and baseline models) in different depth layers (5 to 500 m, 14 layers) and months in 2018 test period. (a) and (b) RMSE and MAPE error at depths in TP area. (c) and (d) RMSE and MAPE error at depths in NWP area. (e) and (f) RMSE and MAPE error at months in TP area. (g) and (h) RMSE and MAPE error at months in NWP area.

model's spatial structures generally align well with observed values across most depths. At depths of 400–500 m, some finer details of the temperature signals are less pronounced, but the overall spatial pattern remains consistent with ARGO data, offering improved accuracy compared to CNN (see Fig. 7). Overall, the predicted temperature distribution is in strong agreement with observed measurements.

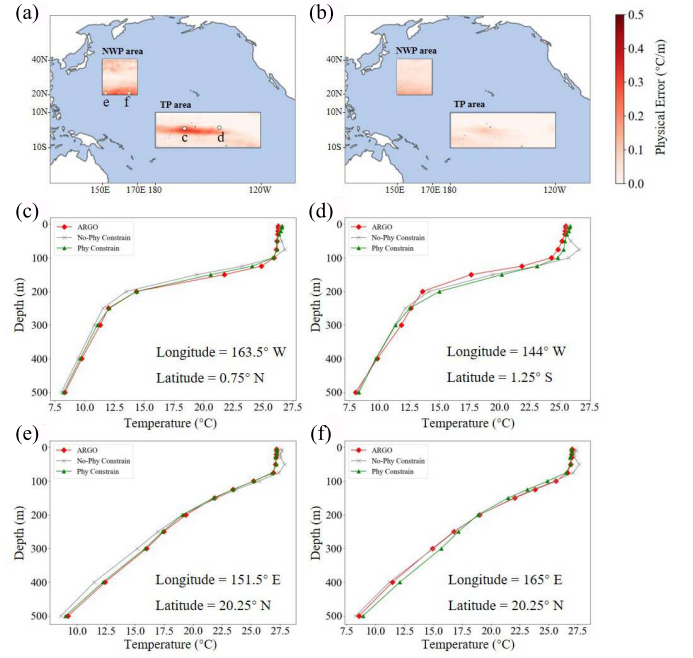


Fig. 6. Spatial distributions of ST predictions at 14 depth layers in (a1)–(n1) TP area and (a2)–(n2) NWP area, respectively. (Left side) IPRC ARGO data, while (Right side) results predicted by 3DMOPGCBANN.

C. Substantial Effectiveness of Integrating Physics Constraints

In this section, we examine the influence of the physics-guided and multiloss optimization modules on predictive performance. The analysis employs physical error distribution and depth profiles, as illustrated in Fig. 8. Physical error is quantified by the residuals of the temperature gradient equation at each grid, aggregated at depth and time dimensions in each spatial location. In the TP area, deep learning models without physical constraints exhibit concentrated physical errors in the equatorial zone. In NWP area, the error displays a distinct latitudinal pattern: in the 30°N–40°N range, errors are widespread but relatively small, while in the 20°N–25°N range, errors are notably larger, possibly linked to the latitudinal distribution of the Kuroshio Extension. With respect to depth, physical errors predominantly occur in shallow waters above 100 m (see Fig. 9). A comparison of Fig. 8(a) and (b) reveals a 45% reduction in physical errors after applying the physics constraint. Furthermore, depth profile analysis demonstrates that the model without prior knowledge constraint exhibits temperature inversions above 200 m [Fig. 8(c)–(f)], violating ocean stratification principles. The 3D-MOPGCBANN model mitigates this issue, significantly improving ST predictions through the integration of prior knowledge and optimization techniques.

The performance of the MO module, evaluated through the KKT theory, optimizes the model from both training cost and performance perspectives. Unlike the previous training approach, which integrates physical knowledge into the neural network loss function with a weight step search ((7)), where w_1 is fixed at 1 and w_2 is searched, the MO module significantly accelerates training and automates model optimization. It achieves Pareto-optimal results in a single training session,

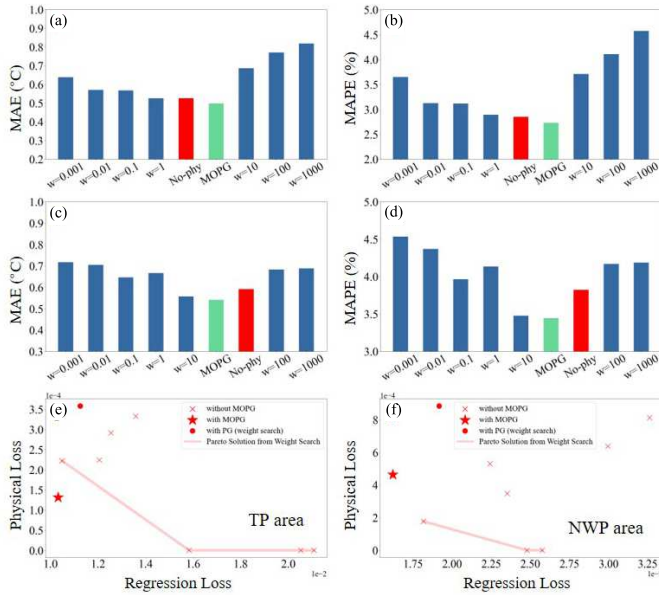


Fig. 7. Spatial distributions of subsurface temperature of ARGO data and CNN predictions at 14 depth layers. The results are ordered by ARGO data and CNN predictions in each subplot.

eliminating the need for iterative experiments. In terms of training efficiency, it reduces overall training time by more than 70%. Furthermore, the module yields superior Pareto solutions, as evidenced by improvements in model accuracy (Table II). As shown in Fig. 10(a)–(d), MAE and MAPE metrics for models without physics constraints, weight search PG models, and MO-PG models demonstrate that the MO-PG models achieve the best results across both TP and NWP areas. Additionally, the module effectively balances data loss and physical loss, reaching the Pareto frontier. The Pareto solutions for both areas [Fig. 10(e) and (f)] show that the MO-PG solutions dominate those of the baseline models, offering better performance in both regression and physical loss. Moreover, solutions with a higher physical weight also benefit from superior accuracy metrics when utilizing the MO-PG approach. Therefore, the MO-PG module optimally balances dual loss functions, achieving the best overall performance.

D. Module Contributions in Enhancing ST Predictions

In Sections V-A–V-C, the effectiveness and practicality of the proposed model are extensively examined and validated through empirical analysis. This segment delineates a suite of ablation studies aimed at discerning the contribution and importance of individual modules. This objective is accomplished by systematically omitting or modifying principal components and observing the repercussions on performance metrics, thereby augmenting understanding of the proposed method. The ablation experimentation is categorized into three configurations: 1) models without the PG and MO modules; 2) models with PG module but without MO module, where the MO module is replaced with a weight-step search strategy, spanning a search domain from 10^k (where k is an integer ranging from -3 to 3); and 3) models with both PG and MO modules, where the MO module refines the PG module by leveraging a multiobjective loss function. For fairness in

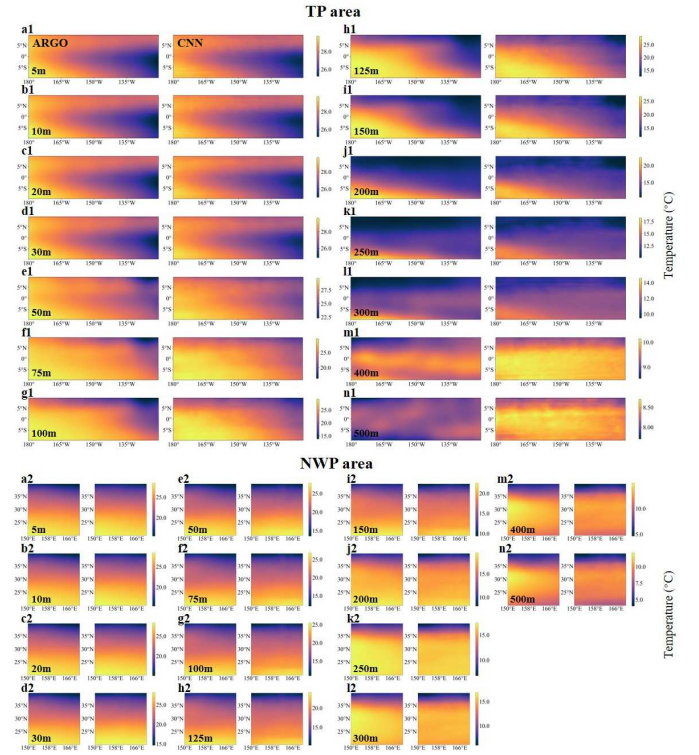


TABLE III
EVALUATION OF ABLATION EXPERIMENTS

Model	Module			RMSE/°C	MAE/°C	MAPE/%	R ²
	CBAM	PG	MO				
TP area	×	×	×	1.164	0.693	3.943	0.978
	✓	×	×	0.940	0.608	3.340	0.985
	×	✓	×	1.156	0.688	3.904	0.978
	×	✓	✓	1.162	0.695	3.937	0.978
	✓	✓	×	0.858	0.568	3.118	0.988
	✓	✓	✓	0.794	0.498	2.733	0.990
	×	×	×	1.243	0.933	6.113	0.952
NWP area	✓	×	×	0.845	0.640	4.014	0.978
	×	✓	×	1.238	0.928	6.075	0.952
	×	✓	✓	1.226	0.924	5.959	0.953
	✓	✓	×	<u>0.759</u>	<u>0.591</u>	<u>3.735</u>	<u>0.982</u>
	✓	✓	✓	0.712	0.542	3.447	0.984
	×	×	×	1.243	0.933	6.113	0.952
	✓	×	×	0.845	0.640	4.014	0.978

Table Note: When solely utilizing the PG module, training is conducted via a weighted stepwise search approach, where $LOSS_{total} = LOSS_{regression} + w * LOSS_{physics}$. w in table denotes the best weight for the physical loss. Detailed evaluation of experiments is shown in Figure 7.

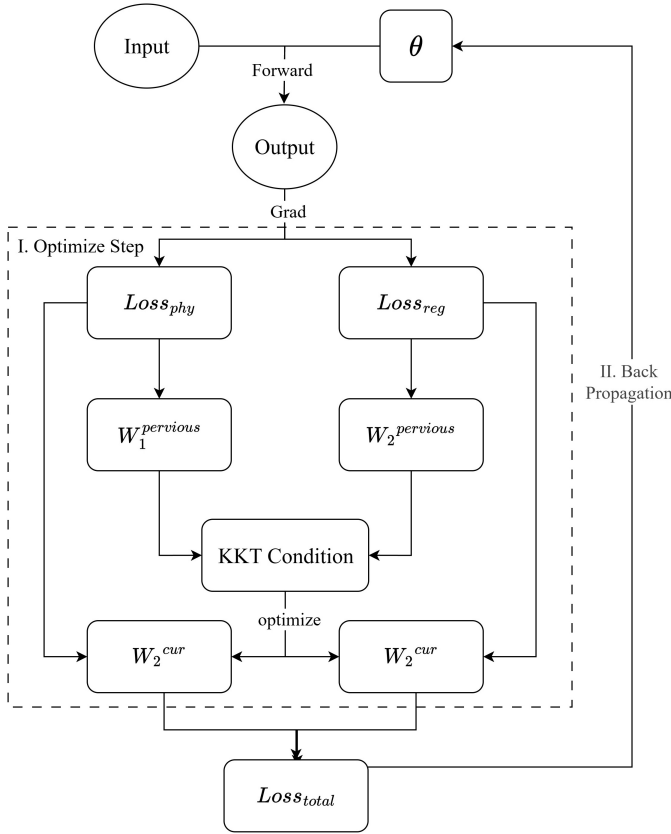


Fig. 10. Comparison on no physics constraint models, weights search PG models and MO-PG models for both TP and NWP areas. (a) and (b) MAE and MAPE error on TP area; (c) and (d) MAE and MAPE error on NWP area; (e) and (f) scatter of regression loss and physical loss of 3 types of models for TP and NWP areas, respectively.

network architectures and hyperparameters maintained across different models.

The quantitative results reveal that models without PG and MO modules are underperforming, as shown in Table III. Models solely relying on CNN architectures without embed-

ding physical knowledge are predisposed to predictions that potentially deviate from prior laws. While models incorporating the PG module exhibit superior performance, second only to the full modules. This underscores the enhancement in ST prediction accuracy attributable to the integration of ocean stratification knowledge, primarily credited to the augmented physical guide instilled by the PG module. Furthermore, experimental findings underscore the pivotal role of the MO module in optimizing the balance between dual loss functions. The complete method, embodying both PG and MO modules, manifests optimal performance, corroborating antecedent observations. Among these, the 3D-MOPGCBANN model demonstrates the most substantial improvement, with RMSE, MAE, MAPE, and R^2 increments of 31.8%, 28.1%, 30.7%, and 1.2% in the TP area, and 42.7%, 41.9%, 43.6%, and 3.4% in the NWP area, respectively. This enhancement is primarily attributed to the modules' capability to effectively amalgamate data information with prior physical laws, thus alleviating inconsistencies in physical prediction and enhancing model accuracy.

VI. CONCLUSION

In this study, we propose a novel 3D-MOPGCBANN neural network to improve the precision of ST prediction. A physics-guided module for ST prediction is designed, which allows the model to learn marine physics more effectively by integrating the model with prior physical knowledge. Spatial and channel attention mechanisms are incorporated within the convolutional neural block, enabling the model to focus more on key features, thereby capturing variations in temperature distributions more accurately. A MO module is proposed, applying the KKT condition to optimize dual loss functions within the physical neural network, to improve overall model performance. Extensive experimentation based on the IPRC ARGO sea temperature dataset demonstrates the efficacy and practicality of the proposed method. The metrics RMSE, MAE, and MAPE for ST prediction achieved

by 3D-MOPGCBANN improve accuracy by 31.8%, 28.1%, and 30.7% in the TP area, and 42.7%, 41.9%, and 43.6% in the NWP area, respectively. The experimental results confirm that models based on the proposed method outperform comparison models in ST prediction tasks.

Beyond the reported gains, the main limitations and potential mitigation methods of this work are as follows. First, the existing model structure is only suitable for single ocean parameter reconstruction and may not fully capture the nonlinear relationships between multiple parameters. In the future, methods such as multitask learning and region clustering learning (such as CGKDN) can be explored to construct coupled models that can simultaneously reconstruct multiple elements. Second, there may still be abnormal phenomena that do not match the actual situation after adding physical constraints to the model under some conditions. The reason may be that prior knowledge is not fully applicable in certain specific regions or conditions, or that the model has limitations in expressing prior knowledge. Enhancing the universality of embedded knowledge is a key issue for future research. Research can be conducted on how to dynamically adjust the expression of prior knowledge to adapt to the changes in different regions or conditions in the following work. Third, although current dual-drive deep learning models can effectively integrate data and prior knowledge, their internal mechanisms still have a “black-box” characteristic and lack interpretability for the model’s decision-making process. Techniques such as attention mechanism visualization, gradient weighted class activation mapping, Shapley value analysis, etc., can be introduced to understand how models use data and prior knowledge for reconstruction, thereby improving the credibility and practicality of the models.

In general, this research presents a highly promising approach for predicting ST, validates the effectiveness of embedding physical knowledge into models, and designs a general learning method of physics guide training and optimization. It offers new perspectives and methodologies for ST prediction. In future work, we will further extend the research scale to larger and deeper ocean regions, look at other ocean parameters by combining the corresponding physical knowledge and ocean equations, and improve the performance of deep learning models.

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